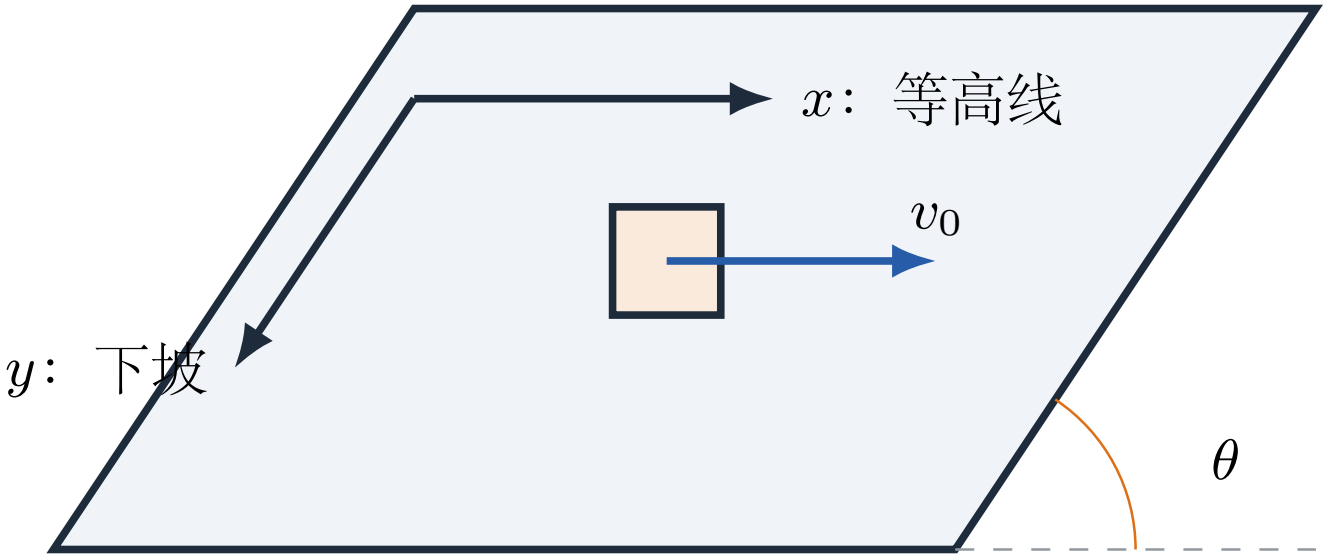
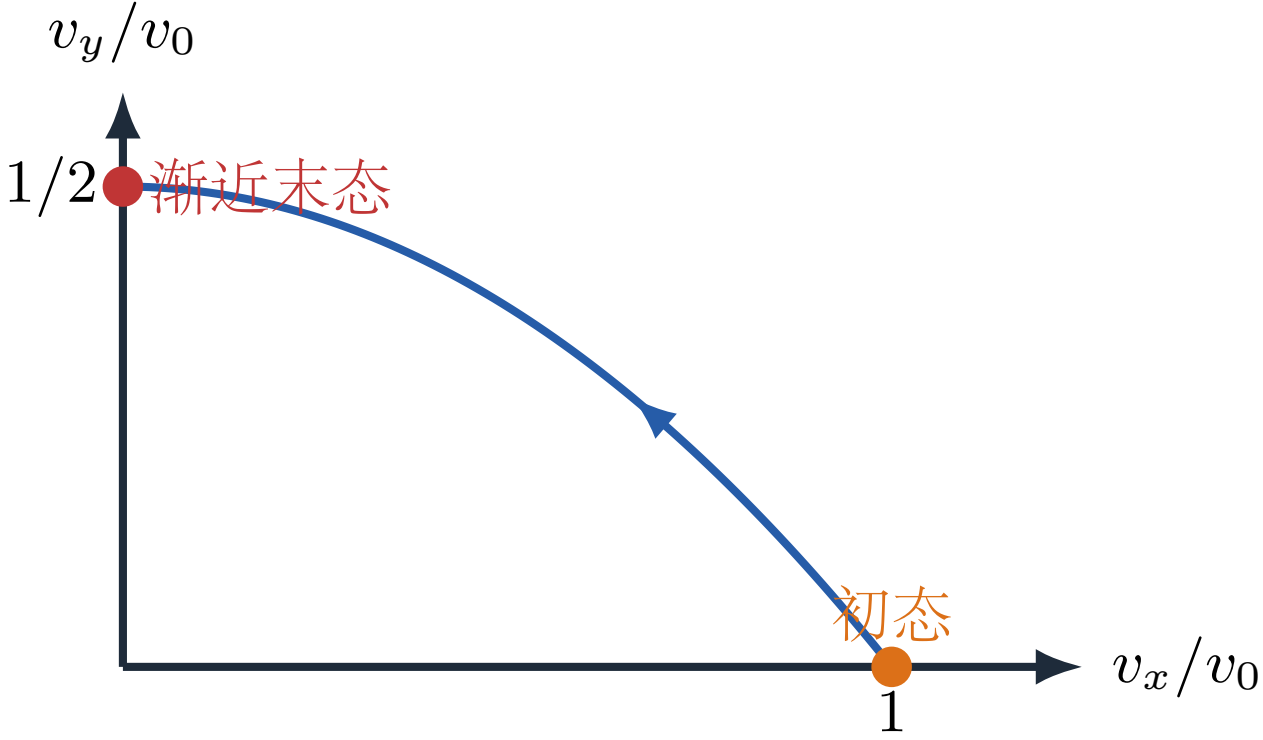


$$\mu = \tan\theta$$

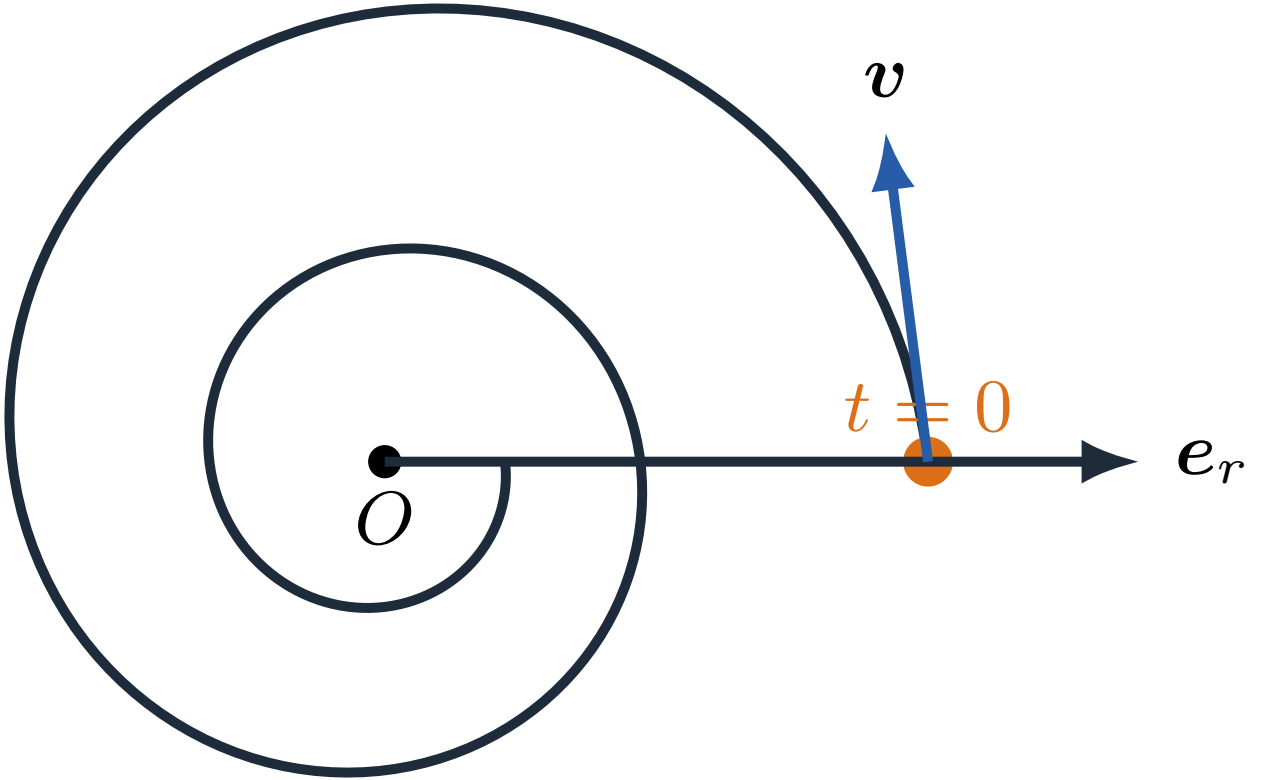
$$N = mg\cos\theta$$



$$v + v_y = v_0$$

$$v_x^2 = v_0^2 - 2v_0 v_y$$

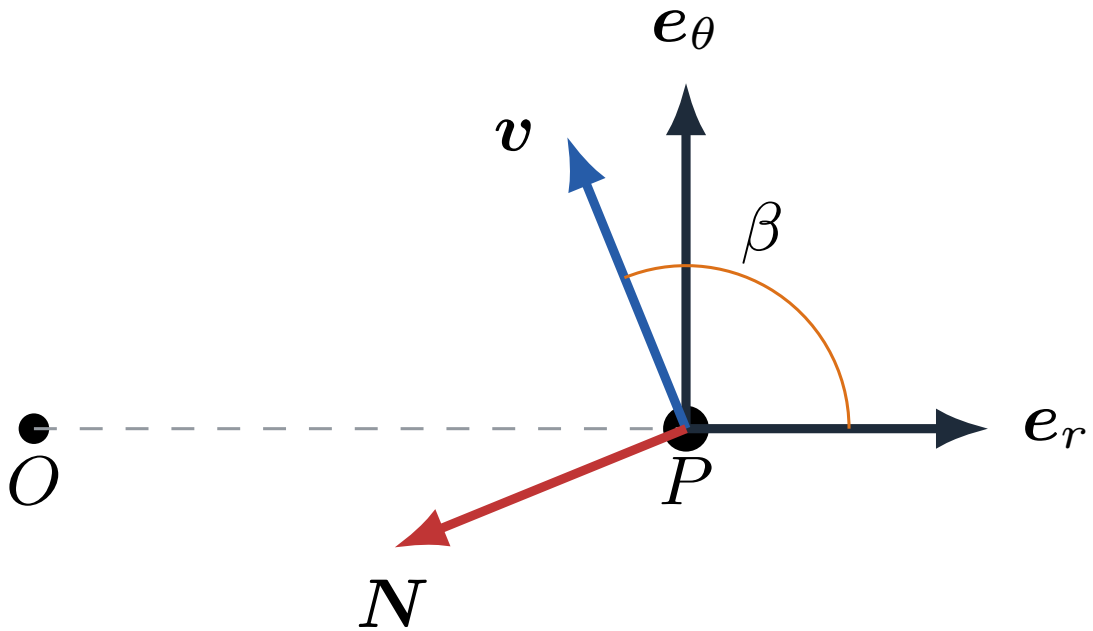
这是速度平面的曲线



$r = r_0 e^{-a\theta}$

θ 增大：向内

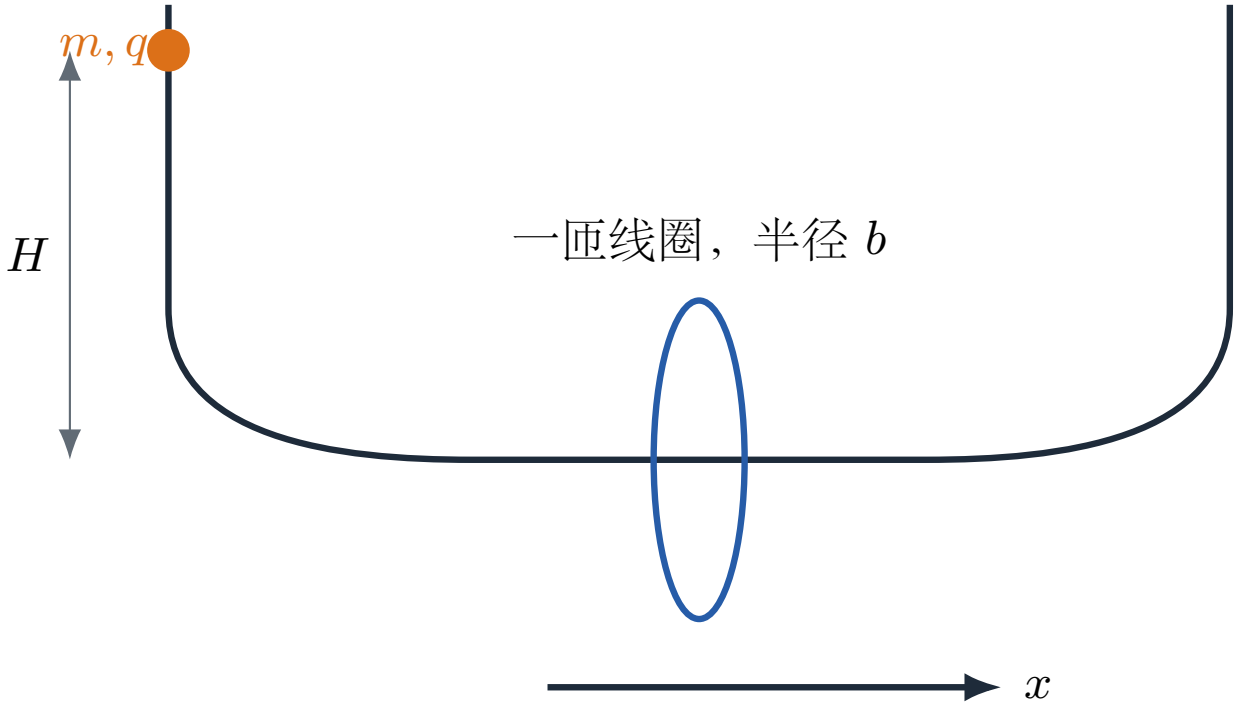
$v = v_0$



$$\boldsymbol{v} \parallel -a\boldsymbol{e}_r + \boldsymbol{e}_\theta$$

$$\boldsymbol{N} \parallel -\boldsymbol{e}_r - a\boldsymbol{e}_\theta$$

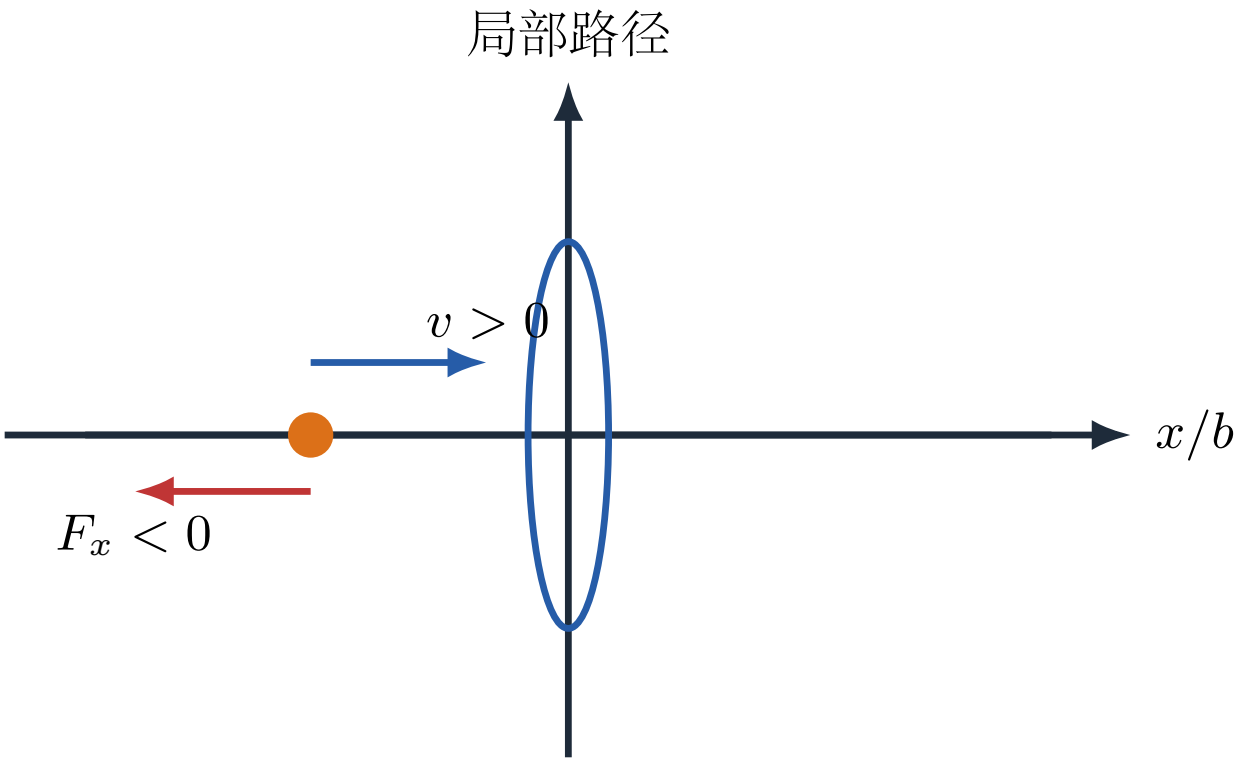
$$\dot{L}_z = -amv_0^2/(1+a^2)$$



$a \ll b$

$R = 2\rho b/a^2$

仅线圈附近近似直轨道

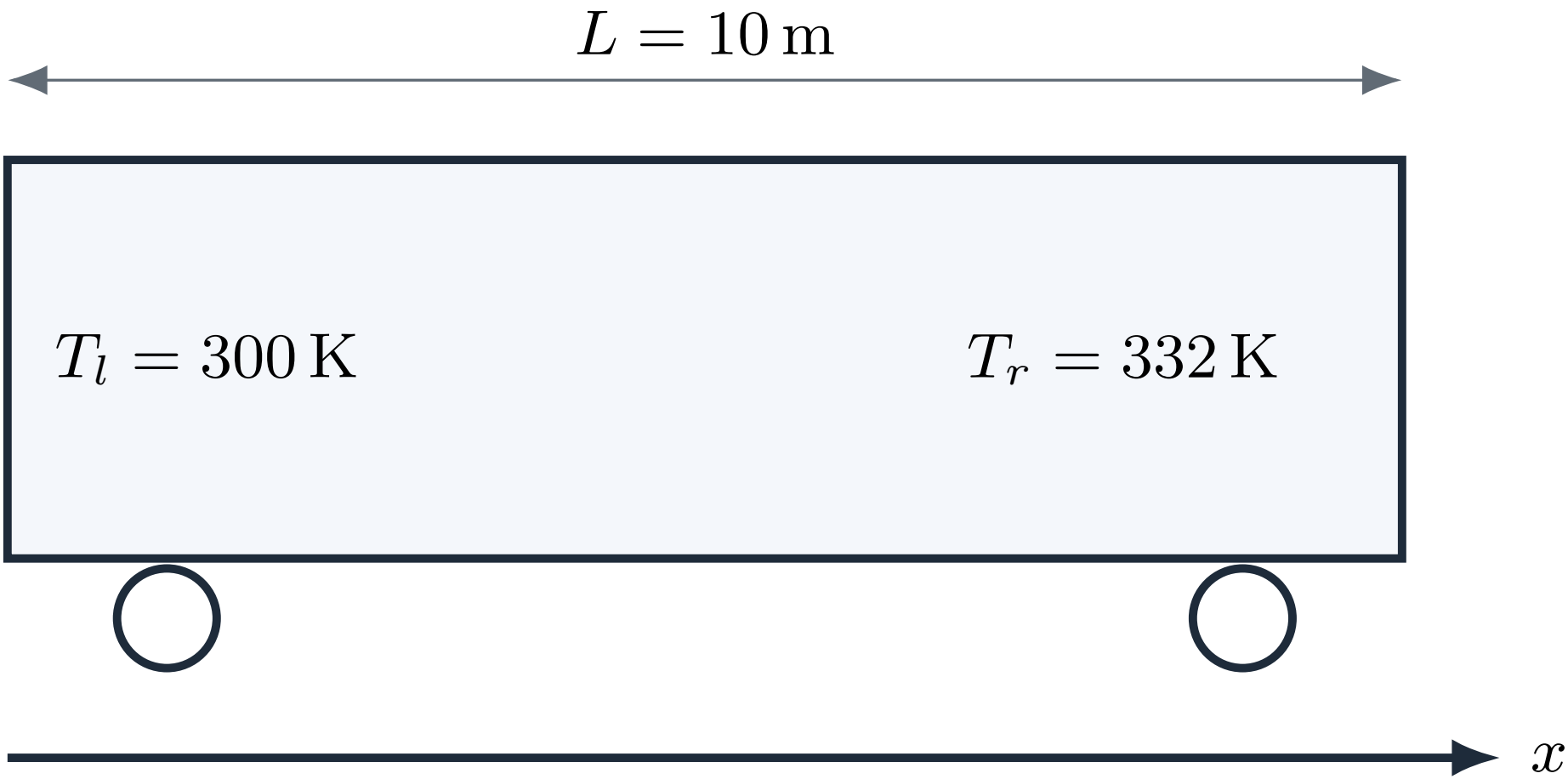


$$F_x = -K^2 v/[R(x^2 + b^2)^3]$$

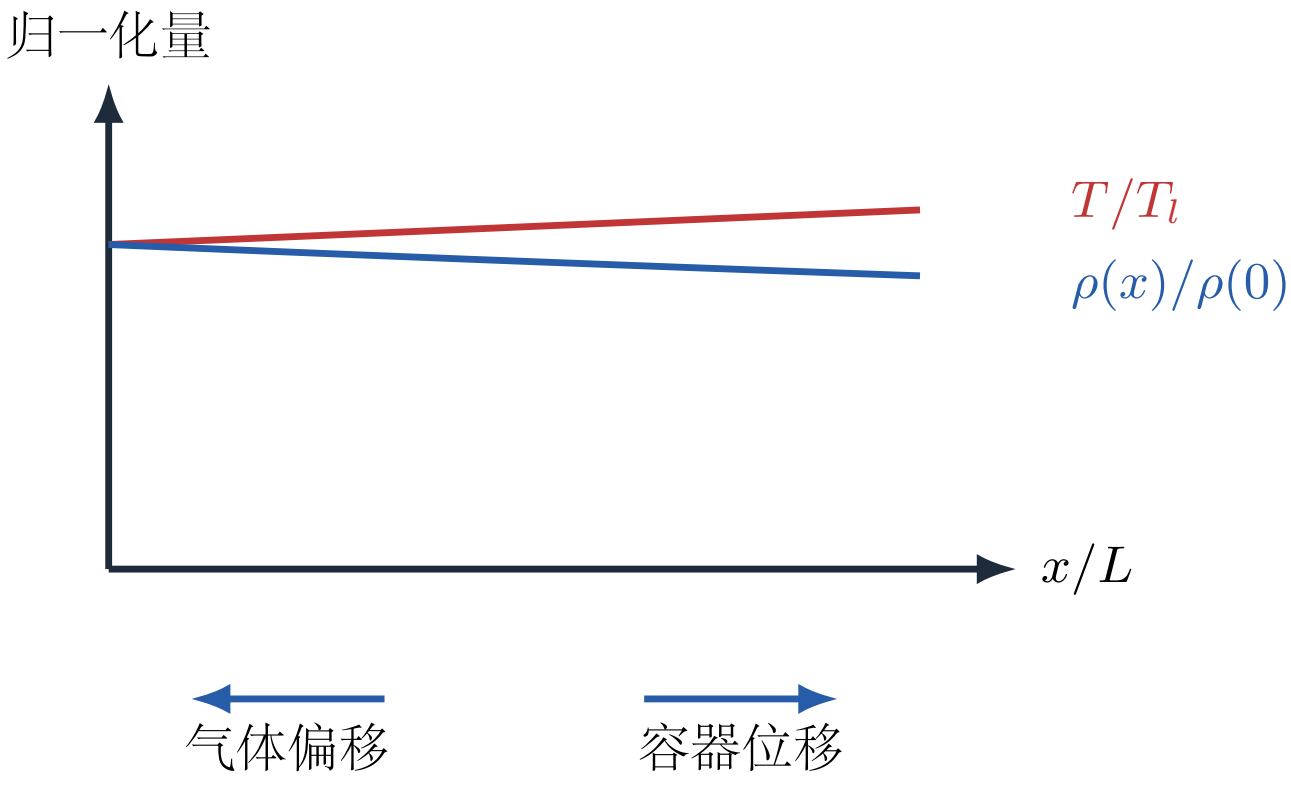
$$\Delta v = 3\pi\mu_0^2 q^2/(32mRb)$$

完整通过要求 $v_{\text{in}} > \Delta v$

$$K = \mu_0 q b^2/2$$



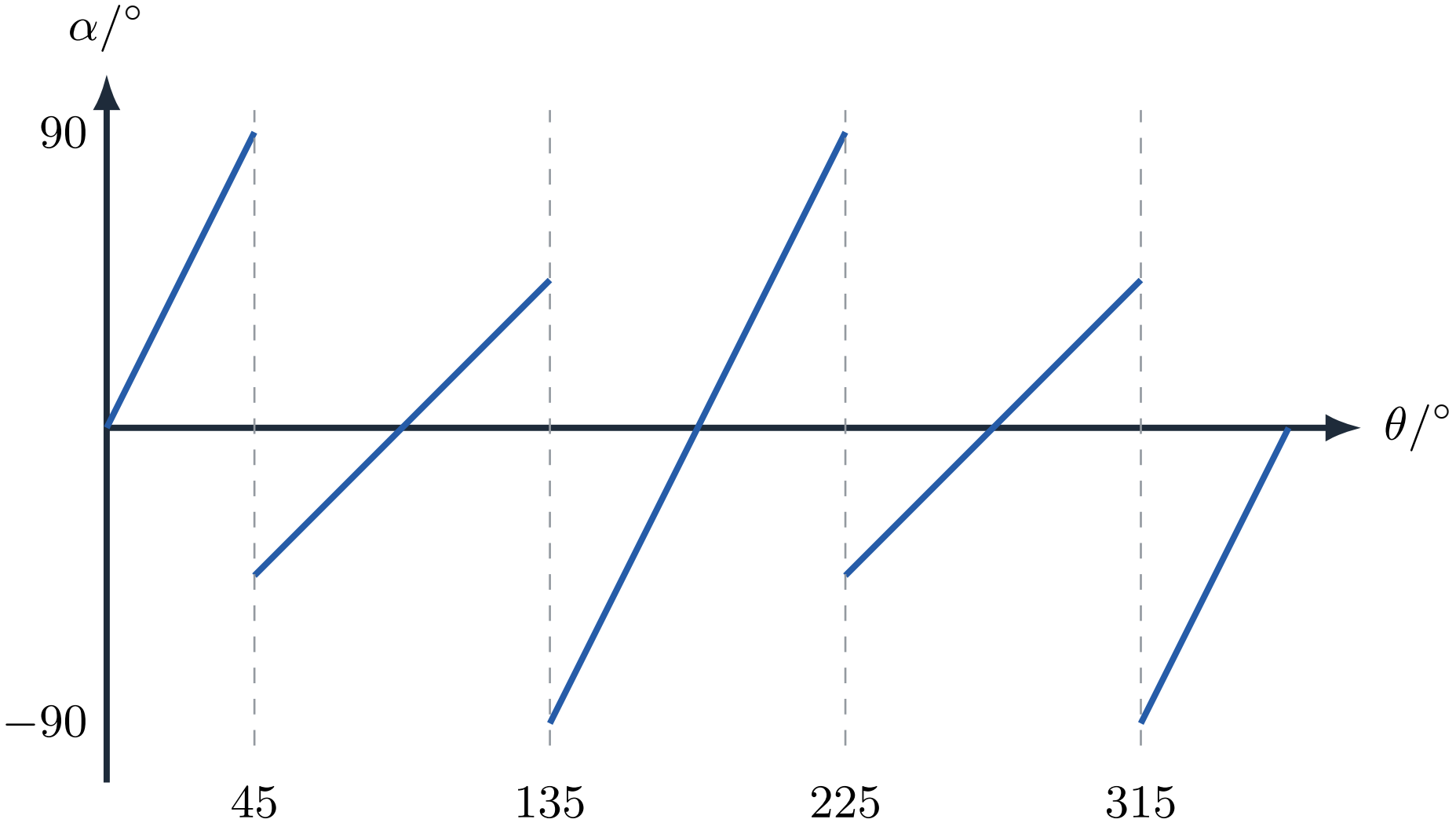
容器与车 $M = 150\,\mathrm{kg}$，气体 $m = 120\,\mathrm{kg}$



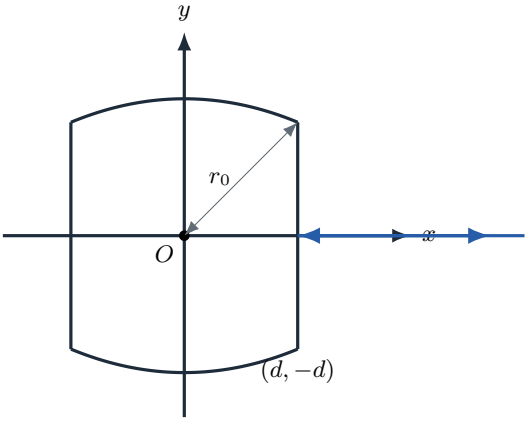
气体相对容器：向左

容器位移 X：向右

$P = 157.865\,\mathrm{atm}$



原扫描折线的关键断点与有向角

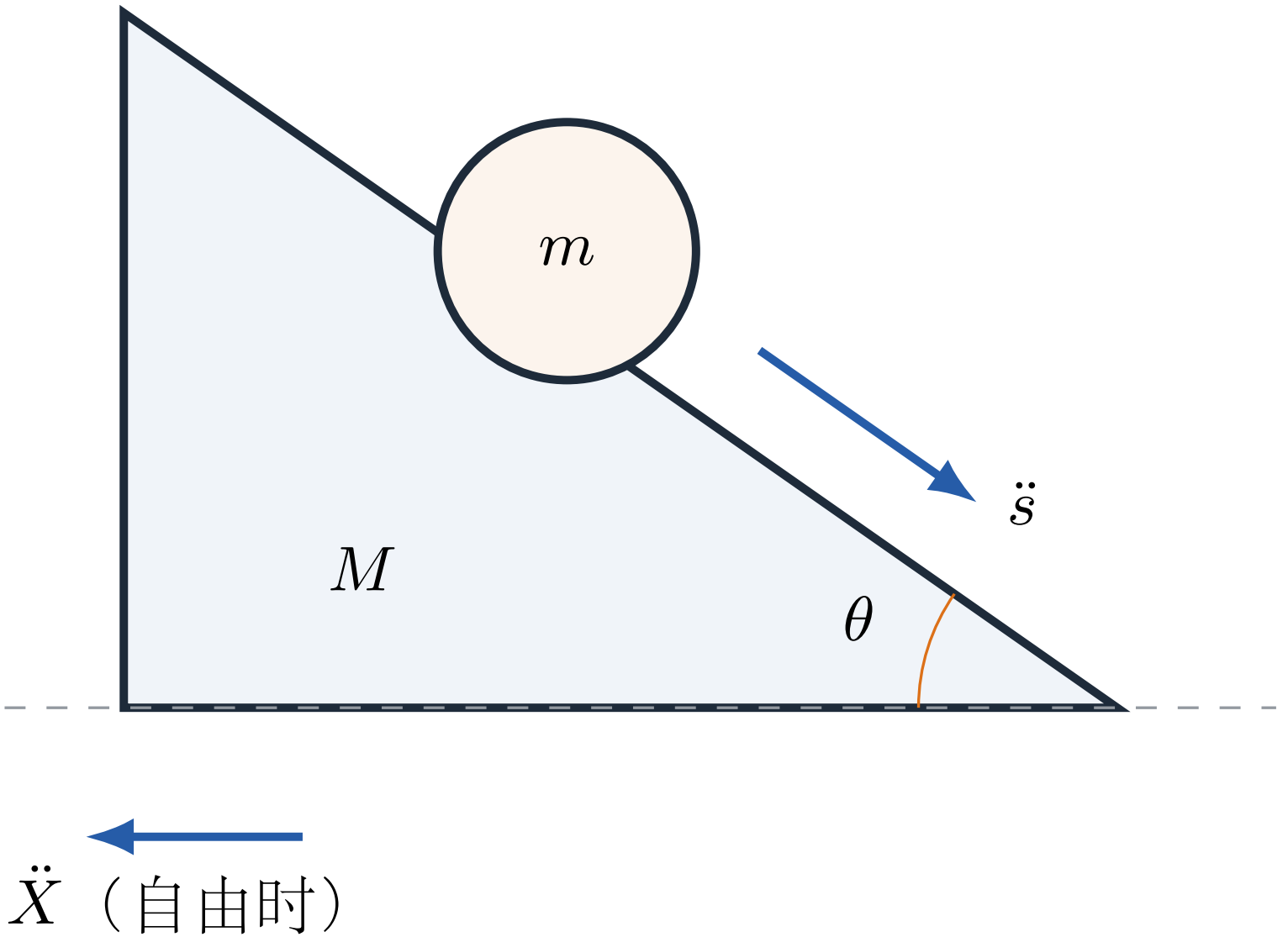


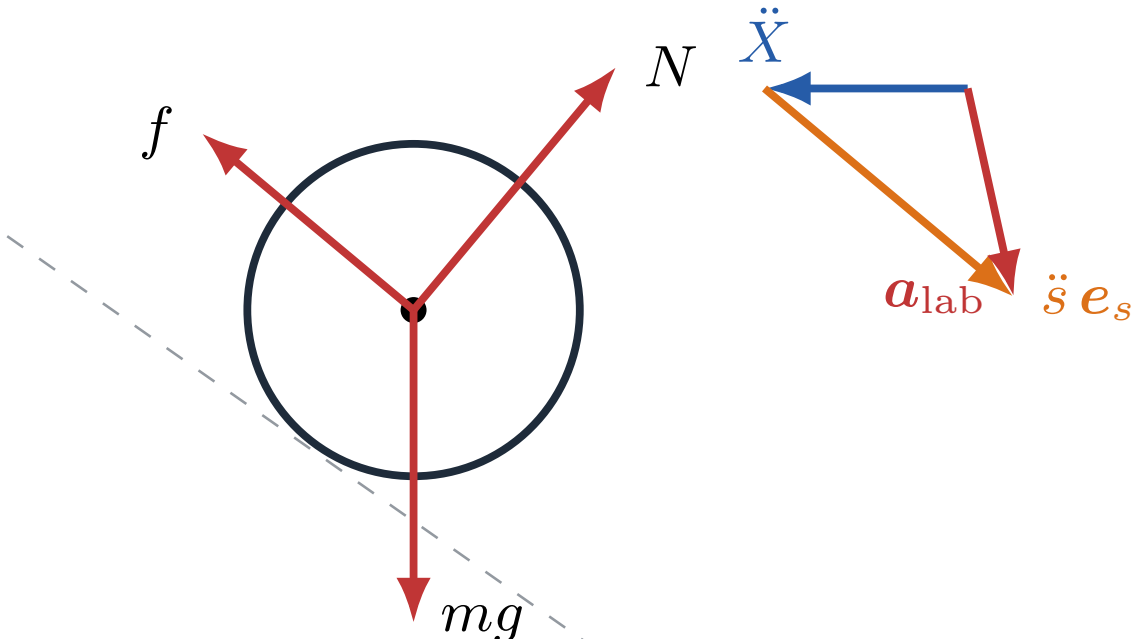
$d = r_0/\sqrt{2}$

$K = (1+\sqrt{2})d$

上帽：$y = (K^2 - x^2)/(2K)$

下帽：$y = -(K^2 - x^2)/(2K)$

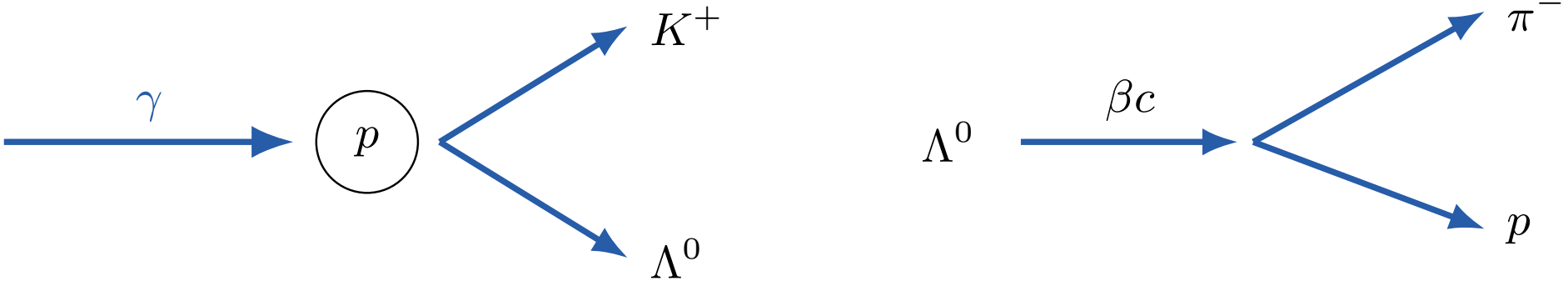

m
M
θ
$\ddot{s}$
$\ddot{X}$（自由时）



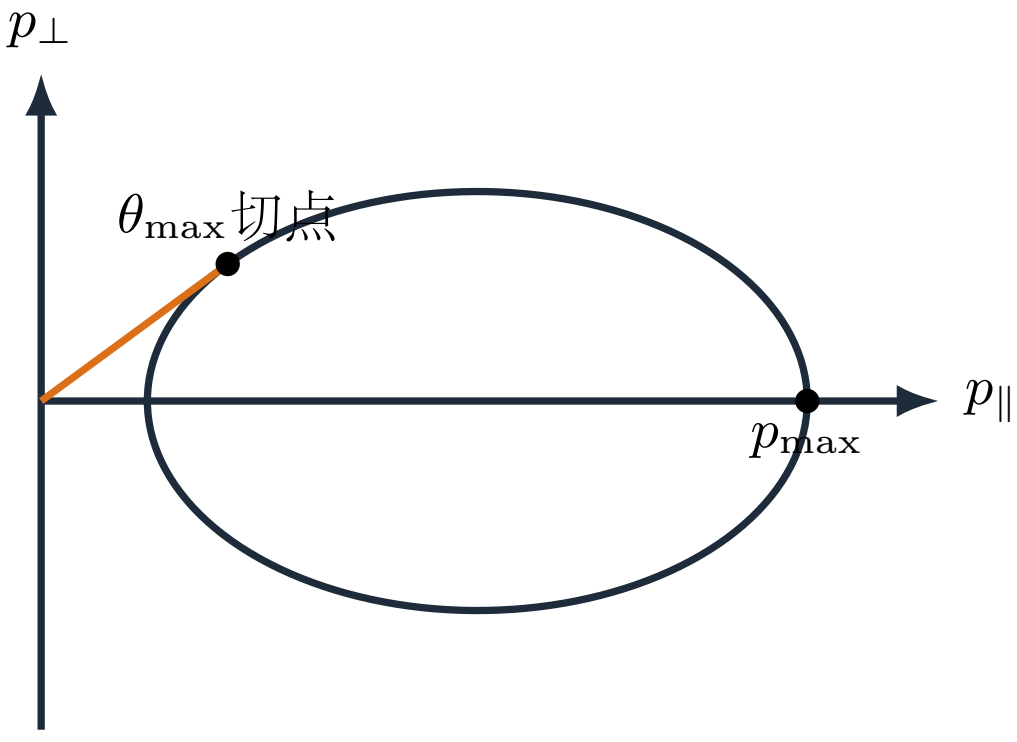
$$\boldsymbol{a}_{\mathrm{lab}} = \ddot{X}\boldsymbol{e}_x + \ddot{s}\boldsymbol{e}_s$$

$$f = \kappa m \ddot{s}$$

$$N = m(g\cos\theta + \ddot{X}\sin\theta)$$



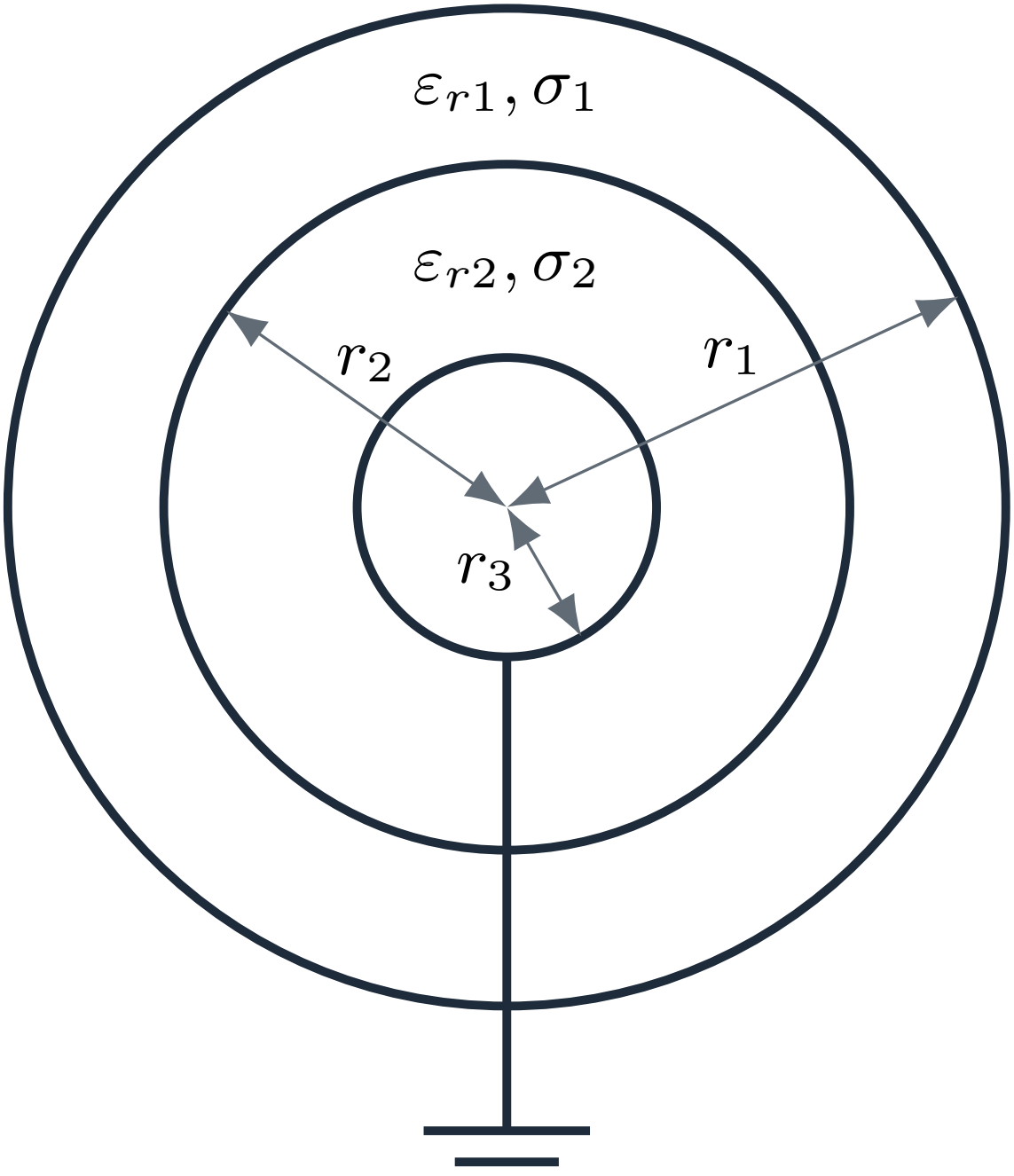
反应阈值与飞行母粒子的两体衰变



$p_{\max} = 369.376\,\mathrm{MeV}/c$

$\theta_{\max} = 36.274°$

最大动量与最大方向角位置不同

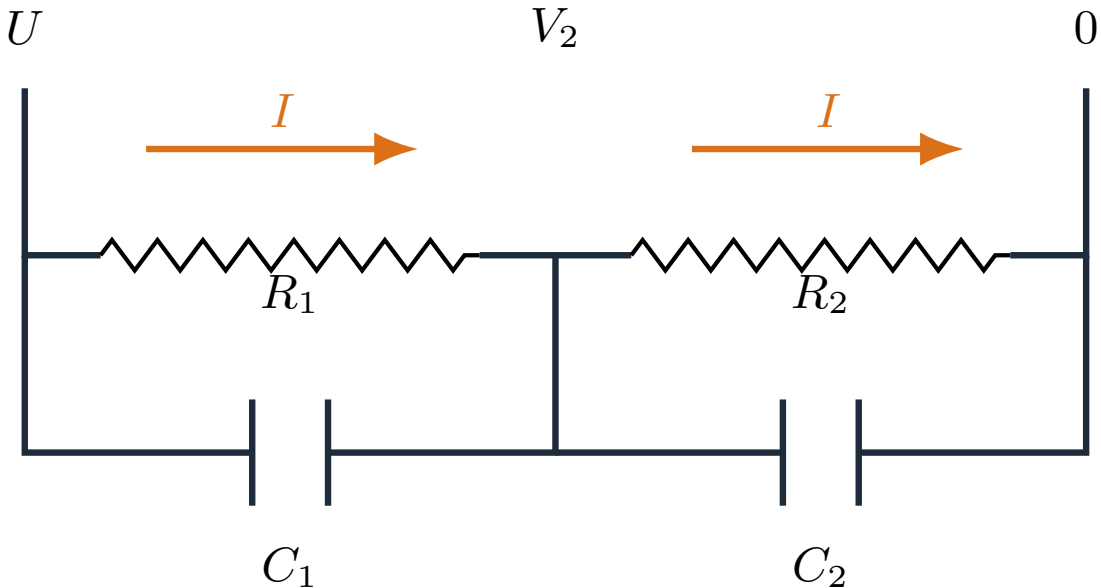


外壳：$U > 0$

中壳：悬浮

内壳：接地

电流径向向内



$$q_{1,\text{inner}} = \varepsilon_0 I/\sigma_1$$

$$q_{f2} = \varepsilon_0 I(\varepsilon_{r2}/\sigma_2 - \varepsilon_{r1}/\sigma_1)$$

稳态由电阻分压决定